

Debajyoti Das

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PROFESSIONAL SUMMARY

Software Engineer (GenAI Engineer) and Data Science enthusiast with 2+ years of experience building scalable backend systems and AI-driven platforms. Skilled in machine learning, NLP, transformer architectures, and generative AI, with hands-on experience using FastAPI and AWS (Lambda, SQS, RDS, Bedrock) to deliver production-grade solutions. Led multiple GenAI and ML initiatives including clinical document automation and conversion rate optimization models, enabling 100+ concurrent users and reducing manual effort by up to 90%.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL (Basic)

Libraries & Frameworks: Pandas, NumPy, Seaborn, Scikit-learn, TensorFlow, PyTorch, FastAPI, LangChain, LangGraph, CrewAI

Cloud & DevOps: AWS (Lambda, S3, SQS, Bedrock, EC2, RDS, ECR, Secrets Manager, Parameter Store)

Developer Tools: VS Code, Git, GitHub, Jupyter Notebook, Power BI Desktop (Basic)

Focus Areas: Generative AI, Machine Learning, Data Science, RAG, Prompt Engineering, LLM Orchestration

Soft Skills: Problem Solving, Adaptability, Leadership, Communication, Time Management

WORK EXPERIENCE

UsefulBI Corporation | Software Engineer I | February 2024 – Present Bangalore, India (Onsite)

Project – Clinical Trial Document Automation Platform | Aug 2024 – Aug 2025

- Built an **end-to-end GenAI platform** for automated generation of **ICF** and **PLPS** clinical trial documents.
- Designed scalable backend services using **FastAPI**, **AWS Lambda**, **SQS**, and **PostgreSQL (RDS)**, supporting **100+ concurrent users**.
- Developed an initial **RAG-based PoC** using **Amazon Titan embeddings**, **OpenSearch**, and **Claude Sonnet 3.5** for accurate document grounding.
- Optimized the pilot by transitioning to an **in-context learning approach** with an **LLM-driven Table-of-Contents agent**, reducing embedding costs and cutting manual medical writing effort by **90%**.
- Productionized the platform using a **Lambda + SQS-based architecture** to handle high user concurrency, with **multi-region selection** and **multi-model orchestration** (Claude Sonnet 3.5/3.7/4.5, Llama-4) for reliable and scalable inference.

Project – Conversion Rate Optimization | Feb 2024 – Aug 2024

- Developed a **machine learning classification model** to predict student likelihood of converting to a **paid subscription after a 21-day trial**, leveraging behavioral engagement features such as platform activity, course consumption, and session frequency.
- Enabled **conversion rate optimization** by identifying **at-risk users** and driving **targeted outreach strategies** (reminders, engagement nudges, and offers), improving trial-to-paid conversion through data-driven student segmentation.

Project – Context-Aware NLQ-to-SQL System | Oct 2025 – Present

- Built a **RAG-based NLQ-to-SQL system** enabling users to query enterprise data using natural language, leveraging **Claude Sonnet 3.7** with **Cognos data models** to provide **business-level context** beyond raw database schemas.
- Implemented **dynamic database/schema selection** and **conversational query refinement** using chat memory, along with **GenAI-powered HTML KPI dashboards and charts** for instant analytics and improved usability for non-technical users.

EDUCATION

National Institute of Technology Kurukshetra, Haryana

Master of Computer Applications

Oct.2021– July 2024

CGPA 8.7

Jadavpur University, West Bengal, India

Bachelor of Science(B.Sc.) in Mathematics (Honours)

July. 2018 – Aug 2021

Percentage: 75.11

Fatepur Srinath Institution, West Bengal, India

WBCHSE, 12th

2018

Percentage: 91.60